Questions

1. **(a)** Prove by induction that, for all integers $n \geq 1$

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

[4]

(b) Hence, or otherwise, write down a factorised expression for the sum of the first $2n$ cubes

$$1^3 + 2^3 + 3^3 + \cdots + (2n)^3$$

[1]

(c) Use the formula in part (a) to write down a factorised expression for the sum of the first n even cubes

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3$$

[2]

(d) Hence, or otherwise, show that

$$1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = an^2(bn^2 - 1)$$

where a and b are rational numbers to be determined. **[3]**

Solution

(a) We prove the result by mathematical induction.

Base Case: When $n = 1$,

$$\sum_{r=1}^{1} r^3 = 1^3 = 1$$

and

$$\frac{1}{4}(1)^2(1+1)^2 = \frac{1}{4} \cdot 1 \cdot 4 = 1$$

So the formula is true for $n = 1$.

Inductive Hypothesis: Assume that for some integer $k \geq 1$,

$$\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$$

Inductive Step: Consider $n = k + 1$.

$$\begin{aligned}
\sum_{r=1}^{k+1} r^3 &= \sum_{r=1}^{k} r^3 + (k+1)^3 \\
&= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \\
&= (k+1)^2\left(\frac{k^2}{4} + k + 1\right) \\
&= (k+1)^2\left(\frac{k^2 + 4k + 4}{4}\right) \\
&= \frac{1}{4}(k+1)^2(k+2)^2
\end{aligned}$$

This is exactly the required formula with $n = k + 1$.

Therefore, if the result is true for $n = k$, it is true for $n = k + 1$.

Since it is true for $n = 1$, it follows by induction that for all integers $n \geq 1$,

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

(b) Using part (a) with n replaced by $2n$,

$$\begin{aligned} 1^3 + 2^3 + 3^3 + \cdots + (2n)^3 &= \sum_{r=1}^{2n} r^3 \\ &= \frac{1}{4}(2n)^2(2n+1)^2 \\ &= n^2(2n+1)^2 \end{aligned}$$

So the factorised expression is

$$n^2(2n+1)^2$$

(c) The first n even cubes are

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3 = \sum_{r=1}^{n}(2r)^3$$

Hence

$$\begin{aligned} \sum_{r=1}^{n}(2r)^3 &= 8\sum_{r=1}^{n} r^3 \\ &= 8\left(\frac{1}{4}n^2(n+1)^2\right) \\ &= 2n^2(n+1)^2 \end{aligned}$$

So the factorised expression is

$$2n^2(n+1)^2$$

(d) The odd cubes are found by subtracting the even cubes from the first $2n$ cubes:

$$\begin{aligned} 1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 &= \left(1^3 + 2^3 + \cdots + (2n)^3\right) - \left(2^3 + 4^3 + \cdots + (2n)^3\right) \\ &= n^2(2n+1)^2 - 2n^2(n+1)^2 \\ &= n^2\left((2n+1)^2 - 2(n+1)^2\right) \\ &= n^2\left((4n^2+4n+1) - (2n^2+4n+2)\right) \\ &= n^2(2n^2-1) \end{aligned}$$

Comparing with

$$an^2(bn^2-1)$$

gives

$$a = 1, \qquad b = 2$$

So

$$1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = n^2(2n^2-1)$$

2. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} r\,2^r = 2 + (n-1)2^{n+1}$$

[5]

Solution

Let

$$P(n) : \sum_{r=1}^{n} r\,2^r = 2 + (n-1)2^{n+1}$$

We will prove that $P(n)$ is true for all $n \geq 1$ by induction.
Base Case: For $n = 1$,

$$\sum_{r=1}^{1} r\,2^r = 1 \cdot 2 = 2$$

Also,

$$2 + (1-1)2^{1+1} = 2 + 0 = 2$$

So the left-hand side equals the right-hand side, and hence $P(1)$ is true.
Inductive Hypothesis: Assume that $P(k)$ is true for some integer $k \geq 1$.
That is, assume

$$\sum_{r=1}^{k} r\,2^r = 2 + (k-1)2^{k+1}$$

Inductive Step: We now prove that $P(k+1)$ is true.
Starting with the left-hand side,

$$\begin{aligned}
\sum_{r=1}^{k+1} r\,2^r &= \sum_{r=1}^{k} r\,2^r + (k+1)2^{k+1} \\
&= \left(2 + (k-1)2^{k+1}\right) + (k+1)2^{k+1} \\
&= 2 + \big((k-1) + (k+1)\big)2^{k+1} \\
&= 2 + 2k\,2^{k+1} \\
&= 2 + k\,2^{k+2} \\
&= 2 + \big((k+1) - 1\big)2^{(k+1)+1}
\end{aligned}$$

This is exactly the required form for $P(k+1)$.
Therefore, if $P(k)$ is true, then $P(k+1)$ is also true.
Conclusion: Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, it follows by mathematical induction that

$$\sum_{r=1}^{n} r\,2^r = 2 + (n-1)2^{n+1}$$

for all $n \geq 1$.

3. Prove by induction that for all positive integers n

$$\sum_{r=1}^{n} \log\left(1 - \frac{1}{4r^2}\right) = \log\left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4}\right)$$ **[6]**

Solution

Let

$$S_n = \sum_{r=1}^{n} \log\left(1 - \frac{1}{4r^2}\right)$$

We prove by induction that

$$S_n = \log\left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4}\right)$$

for all positive integers n.

Base Case: $n = 1$.

$$S_1 = \log\left(1 - \frac{1}{4}\right) = \log\left(\frac{3}{4}\right)$$

Also,

$$\log\left(\frac{(2\cdot 1)!(2\cdot 1+1)!}{2^{4\cdot 1}(1!)^4}\right) = \log\left(\frac{2!\,3!}{2^4}\right) = \log\left(\frac{2\cdot 6}{16}\right) = \log\left(\frac{3}{4}\right)$$

So the result is true when $n = 1$.

Inductive Hypothesis.

Assume that for some positive integer k,

$$S_k = \log\left(\frac{(2k)!(2k+1)!}{2^{4k}(k!)^4}\right)$$

Inductive Step.

We must show that

$$S_{k+1} = \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k+4}((k+1)!)^4}\right)$$

Starting from the definition of S_{k+1},

$$S_{k+1} = S_k + \log\left(1 - \frac{1}{4(k+1)^2}\right)$$

First simplify the extra term:

$$\begin{aligned} 1 - \frac{1}{4(k+1)^2} &= \frac{4(k+1)^2 - 1}{4(k+1)^2} \\ &= \frac{(2k+2)^2 - 1}{(2k+2)^2} \\ &= \frac{(2k+1)(2k+3)}{(2k+2)^2} \end{aligned}$$

So, using the inductive hypothesis and the law $\log a + \log b = \log(ab)$,

$$\begin{aligned} S_{k+1} &= \log\left(\frac{(2k)!(2k+1)!}{2^{4k}(k!)^4}\right) + \log\left(\frac{(2k+1)(2k+3)}{(2k+2)^2}\right) \\ &= \log\left(\frac{(2k)!(2k+1)!(2k+1)(2k+3)}{2^{4k}(k!)^4(2k+2)^2}\right) \end{aligned}$$

Now use the factorial identities

$$(2k+2)! = (2k+2)(2k+1)(2k)!$$

and

$$(2k+3)! = (2k+3)(2k+2)(2k+1)!$$

Multiplying these together gives

$$\begin{aligned}(2k+2)!(2k+3)! &= (2k+2)(2k+1)(2k)!\,(2k+3)(2k+2)(2k+1)! \\ &= (2k+2)^2(2k+1)(2k+3)(2k)!(2k+1)!\end{aligned}$$

Therefore

$$(2k)!(2k+1)!(2k+1)(2k+3) = \frac{(2k+2)!(2k+3)!}{(2k+2)^2}$$

Substituting this into the expression for S_{k+1},

$$S_{k+1} = \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k}(k!)^4(2k+2)^4}\right)$$

Now simplify the denominator:

$$(2k+2)^4 = [2(k+1)]^4 = 2^4(k+1)^4$$

and

$$((k+1)!)^4 = ((k+1)k!)^4 = (k+1)^4(k!)^4$$

Hence

$$2^{4k}(k!)^4(2k+2)^4 = 2^{4k}\cdot 2^4\cdot (k+1)^4(k!)^4 = 2^{4k+4}((k+1)!)^4$$

So

$$S_{k+1} = \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k+4}((k+1)!)^4}\right)$$

This is exactly the required result for $n = k+1$.
Conclusion.
The statement is true for $n = 1$, and if it is true for $n = k$, then it is true for $n = k+1$. Therefore, by mathematical induction,

$$\sum_{r=1}^{n} \log\left(1 - \frac{1}{4r^2}\right) = \log\left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4}\right)$$

for all positive integers n.

4. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$ **[5]**

Solution

We prove the result by induction on n.

Base Case: $n = 1$

The left-hand side is

$$\sum_{r=1}^{1} \frac{1}{r(r+1)(r+2)} = \frac{1}{1 \cdot 2 \cdot 3} = \frac{1}{6}$$

The right-hand side is

$$\left. \frac{n(n+3)}{4(n+1)(n+2)} \right|_{n=1} = \frac{1(1+3)}{4(1+1)(1+2)} = \frac{4}{4 \cdot 2 \cdot 3} = \frac{1}{6}$$

So the formula is true when $n = 1$.

Inductive Hypothesis:

Assume that for some $k \in \mathbb{Z}^+$,

$$\sum_{r=1}^{k} \frac{1}{r(r+1)(r+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$$

Inductive Step:

We now show that the formula is true for $n = k + 1$.

Starting with the left-hand side,

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{1}{r(r+1)(r+2)} &= \sum_{r=1}^{k} \frac{1}{r(r+1)(r+2)} + \frac{1}{(k+1)(k+2)(k+3)} \\
&= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \qquad \text{(by the inductive hypothesis)} \\
&= \frac{k(k+3)(k+3)}{4(k+1)(k+2)(k+3)} + \frac{4}{4(k+1)(k+2)(k+3)} \\
&= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)} \\
&= \frac{k(k^2 + 6k + 9) + 4}{4(k+1)(k+2)(k+3)} \\
&= \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)} \\
&= \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)} \\
&= \frac{(k+1)(k+4)}{4(k+2)(k+3)}
\end{aligned}$$

Now write this in the required form:

$$\frac{(k+1)(k+4)}{4(k+2)(k+3)} = \frac{(k+1)\big((k+1)+3\big)}{4\big((k+1)+1\big)\big((k+1)+2\big)}$$

So the statement is true for $n = k + 1$.

Conclusion:

Since the statement is true for $n = 1$, and true for $n = k + 1$ whenever it is true for $n = k$, it follows by mathematical induction that

$$\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)} \quad \text{for all } n \in \mathbb{Z}^+$$

Hence the result is proved.

5. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} (2^r + 2 \times 3^r) = 2^{n+1} + 3^{n+1} - 5$$

[5]

Solution

We use proof by induction on n.
Base Case: $n = 1$.
The left-hand side is

$$\sum_{r=1}^{1} (2^r + 2 \times 3^r) = 2^1 + 2 \times 3^1 = 2 + 6 = 8$$

The right-hand side is

$$2^{1+1} + 3^{1+1} - 5 = 2^2 + 3^2 - 5 = 4 + 9 - 5 = 8$$

So the result is true when $n = 1$.
Induction Hypothesis:
Assume that for some $k \geq 1$,

$$\sum_{r=1}^{k} (2^r + 2 \times 3^r) = 2^{k+1} + 3^{k+1} - 5$$

Inductive Step: prove the result for $n = k + 1$.
Starting with the left-hand side for $n = k + 1$,

$$\sum_{r=1}^{k+1} (2^r + 2 \times 3^r) = \sum_{r=1}^{k} (2^r + 2 \times 3^r) + \left(2^{k+1} + 2 \times 3^{k+1}\right)$$

Now use the induction hypothesis:

$$\begin{aligned}\sum_{r=1}^{k+1} (2^r + 2 \times 3^r) &= \left(2^{k+1} + 3^{k+1} - 5\right) + \left(2^{k+1} + 2 \times 3^{k+1}\right) \\ &= 2^{k+1} + 2^{k+1} + 3^{k+1} + 2 \cdot 3^{k+1} - 5 \\ &= 2^{k+2} + 3 \cdot 3^{k+1} - 5 \\ &= 2^{k+2} + 3^{k+2} - 5\end{aligned}$$

This is exactly the required result for $n = k + 1$.
Conclusion:
Since the statement is true for $n = 1$, and true for $n = k + 1$ whenever it is true for $n = k$, it follows by induction that

$$\sum_{r=1}^{n} (2^r + 2 \times 3^r) = 2^{n+1} + 3^{n+1} - 5$$

for all $n \geq 1$.

6. Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^{n} r3^{r-1} = \frac{1 + (2n-1)3^n}{4}$$ **[6]**

Solution

Let

$$S_n = \sum_{r=1}^{n} r3^{r-1}$$

We prove by induction that

$$S_n = \frac{1 + (2n-1)3^n}{4}$$

for all $n \in \mathbb{Z}^+$.
Base Case: $n = 1$.
The left-hand side is

$$\sum_{r=1}^{1} r3^{r-1} = 1 \cdot 3^0 = 1$$

The right-hand side is

$$\frac{1 + (2 \cdot 1 - 1)3^1}{4} = \frac{1+3}{4} = 1$$

So the formula is true when $n = 1$.
Induction Hypothesis:
Assume that the statement is true for $n = k$, where $k \in \mathbb{Z}^+$. That is, assume

$$\sum_{r=1}^{k} r3^{r-1} = \frac{1 + (2k-1)3^k}{4}$$

Induction Step:
We must show that the formula is then true for $n = k + 1$.
Starting with the left-hand side for $n = k + 1$,

$$\begin{aligned} \sum_{r=1}^{k+1} r3^{r-1} &= \sum_{r=1}^{k} r3^{r-1} + (k+1)3^k \\ &= \frac{1 + (2k-1)3^k}{4} + (k+1)3^k \end{aligned}$$

using the induction hypothesis.
Now write the second term over a denominator of 4:

$$\begin{aligned} \sum_{r=1}^{k+1} r3^{r-1} &= \frac{1 + (2k-1)3^k}{4} + \frac{4(k+1)3^k}{4} \\ &= \frac{1 + \big((2k-1) + 4k + 4\big)3^k}{4} \\ &= \frac{1 + (6k+3)3^k}{4} \\ &= \frac{1 + 3(2k+1)3^k}{4} \\ &= \frac{1 + (2k+1)3^{k+1}}{4} \end{aligned}$$

But this is exactly the required formula with $n = k + 1$, since

$$\frac{1 + (2(k+1)-1)3^{k+1}}{4} = \frac{1 + (2k+1)3^{k+1}}{4}$$

So, assuming the statement is true for $n = k$, we have shown it is true for $n = k + 1$.

Conclusion:
The statement is true for $n = 1$, and true for $n = k + 1$ whenever it is true for $n = k$. Therefore, by mathematical induction,

$$\sum_{r=1}^{n} r3^{r-1} = \frac{1 + (2n-1)3^n}{4}$$

for all $n \in \mathbb{Z}^+$.

7. **(a)** Prove by induction that, for all positive integers n

$$\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$ **[6]**

(b) Hence, show that, for all positive integers n

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{(an+b)(cn+d)}$$

where a, b, c and d are integers to be determined. **[3]**

Solution

(a) Let

$$S_n = \sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)}$$

We prove by induction that

$$S_n = \frac{n}{2n+1}$$

for all positive integers n.

Base Case: $n = 1$

$$S_1 = \frac{1}{(2 \cdot 1 - 1)(2 \cdot 1 + 1)} = \frac{1}{1 \cdot 3} = \frac{1}{3}$$

and

$$\left.\frac{n}{2n+1}\right|_{n=1} = \frac{1}{2 \cdot 1 + 1} = \frac{1}{3}$$

So the result is true when $n = 1$.

Inductive Hypothesis:

Assume that for some positive integer k,

$$S_k = \sum_{r=1}^{k} \frac{1}{(2r-1)(2r+1)} = \frac{k}{2k+1}$$

Inductive Step:

Consider S_{k+1}:

$$\begin{aligned} S_{k+1} &= \sum_{r=1}^{k+1} \frac{1}{(2r-1)(2r+1)} \\ &= \sum_{r=1}^{k} \frac{1}{(2r-1)(2r+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\ &= S_k + \frac{1}{(2k+1)(2k+3)} \end{aligned}$$

Using the inductive hypothesis,

$$\begin{aligned} S_{k+1} &= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{k(2k+3)+1}{(2k+1)(2k+3)} \\ &= \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\ &= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \\ &= \frac{k+1}{2k+3} \end{aligned}$$

But

$$\frac{k+1}{2k+3} = \frac{k+1}{2(k+1)+1}$$

So the statement is true for $n = k+1$.

Therefore, by induction, for all positive integers n,

$$\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$

Hence proved.

(b) Using part (a),

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \sum_{r=1}^{2n} \frac{1}{(2r-1)(2r+1)} - \sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)}$$

So

$$\begin{aligned} \sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} &= \frac{2n}{2(2n)+1} - \frac{n}{2n+1} \\ &= \frac{2n}{4n+1} - \frac{n}{2n+1} \\ &= \frac{2n(2n+1) - n(4n+1)}{(4n+1)(2n+1)} \\ &= \frac{4n^2+2n-4n^2-n}{(4n+1)(2n+1)} \\ &= \frac{n}{(4n+1)(2n+1)} \end{aligned}$$

Hence

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{(an+b)(cn+d)}$$

with

$$a = 4, \quad b = 1, \quad c = 2, \quad d = 1$$

8. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$$

[5]

Solution

We prove the result by induction.
Base case: $n = 1$.
The left-hand side is

$$\sum_{r=1}^{1} \frac{r}{2^r} = \frac{1}{2}$$

The right-hand side is

$$2 - \frac{1+2}{2^1} = 2 - \frac{3}{2} = \frac{1}{2}$$

So the formula is true when $n = 1$.
Inductive hypothesis: Assume that for some integer $k \geq 1$,

$$\sum_{r=1}^{k} \frac{r}{2^r} = 2 - \frac{k+2}{2^k}$$

Inductive step: We must show that

$$\sum_{r=1}^{k+1} \frac{r}{2^r} = 2 - \frac{(k+1)+2}{2^{k+1}}$$

Starting from the left-hand side,

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{r}{2^r} &= \sum_{r=1}^{k} \frac{r}{2^r} + \frac{k+1}{2^{k+1}} \\
&= \left(2 - \frac{k+2}{2^k}\right) + \frac{k+1}{2^{k+1}} \quad \text{by the inductive hypothesis} \\
&= 2 - \frac{2k+4}{2^{k+1}} + \frac{k+1}{2^{k+1}} \\
&= 2 - \frac{2k+4-(k+1)}{2^{k+1}} \\
&= 2 - \frac{k+3}{2^{k+1}} \\
&= 2 - \frac{(k+1)+2}{2^{k+1}}
\end{aligned}$$

So the statement is true for $n = k+1$.
Conclusion: Since the result is true for $n = 1$, and true for $n = k+1$ whenever it is true for $n = k$, it follows by induction that

$$\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$$

for all integers $n \geq 1$.

9. **(a)** Prove by induction that for all positive integers n

$$\sum_{r=1}^{n}(2r-1)^3 = n^2(2n^2-1)$$

[6]

(b) Given that

$$\sum_{r=1}^{2n}(2r-1)^3 = k\sum_{r=1}^{n}(2r-1)^3$$

show that

$$n^2 = \frac{k-4}{2(k-16)}$$

[5]

Solution

(a) Let

$$S_n = \sum_{r=1}^{n}(2r-1)^3$$

We prove by induction that

$$S_n = n^2(2n^2-1)$$

for all positive integers n.

Base Case: $n = 1$

$$S_1 = (2 \cdot 1 - 1)^3 = 1^3 = 1$$

The formula gives

$$1^2(2 \cdot 1^2 - 1) = 1(2-1) = 1$$

So the result is true when $n = 1$.

Inductive Hypothesis:

Assume that for some positive integer k,

$$S_k = \sum_{r=1}^{k}(2r-1)^3 = k^2(2k^2-1)$$

Inductive Step: Prove the result for $n = k+1$

Starting from S_{k+1},

$$\begin{aligned} S_{k+1} &= \sum_{r=1}^{k+1}(2r-1)^3 \\ &= \sum_{r=1}^{k}(2r-1)^3 + (2(k+1)-1)^3 \\ &= S_k + (2k+1)^3 \end{aligned}$$

Now use the inductive hypothesis:

$$\begin{aligned} S_{k+1} &= k^2(2k^2-1) + (2k+1)^3 \\ &= 2k^4 - k^2 + (8k^3 + 12k^2 + 6k + 1) \\ &= 2k^4 + 8k^3 + 11k^2 + 6k + 1 \end{aligned}$$

We now factor this in the required form:

$$\begin{aligned} (k+1)^2\big(2(k+1)^2 - 1\big) &= (k+1)^2(2k^2+4k+1) \\ &= (k^2+2k+1)(2k^2+4k+1) \\ &= 2k^4 + 8k^3 + 11k^2 + 6k + 1 \end{aligned}$$

Hence

$$S_{k+1} = (k+1)^2\big(2(k+1)^2 - 1\big)$$

So, if the formula is true for $n = k$, it is also true for $n = k + 1$.

Therefore, by mathematical induction,

$$\sum_{r=1}^{n}(2r-1)^3 = n^2(2n^2 - 1)$$

for all positive integers n.

(b) From part (a),

$$\sum_{r=1}^{n}(2r-1)^3 = n^2(2n^2 - 1)$$

So with $2n$ in place of n,

$$\begin{aligned}\sum_{r=1}^{2n}(2r-1)^3 &= (2n)^2\big(2(2n)^2 - 1\big)\\ &= 4n^2(8n^2 - 1)\end{aligned}$$

We are given that

$$\sum_{r=1}^{2n}(2r-1)^3 = k\sum_{r=1}^{n}(2r-1)^3$$

Substitute the formula from part (a):

$$4n^2(8n^2 - 1) = k\big(n^2(2n^2 - 1)\big)$$

Since n is a positive integer, $n^2 \neq 0$, so divide through by n^2:

$$4(8n^2 - 1) = k(2n^2 - 1)$$

Expand:

$$32n^2 - 4 = 2kn^2 - k$$

Rearranging,

$$\begin{aligned}32n^2 - 2kn^2 &= 4 - k\\ (32 - 2k)n^2 &= 4 - k\end{aligned}$$

Multiply top and bottom by -1:

$$(2k - 32)n^2 = k - 4$$

Hence

$$n^2 = \frac{k-4}{2k-32} = \frac{k-4}{2(k-16)}$$

Therefore,

$$n^2 = \frac{k-4}{2(k-16)}$$

10. Prove by induction that for all positive integers n,

$$\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

[6]

Solution

We prove the result by mathematical induction.
We want to show that for all positive integers n,

$$\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

Base Case: $n = 1$
The left-hand side is

$$\sum_{r=1}^{1} r(r+1)(r+2) = 1(2)(3) = 6$$

The right-hand side is

$$\frac{1}{4}(1)(2)(3)(4) = 6$$

So the statement is true when $n = 1$.
Inductive Hypothesis:
Assume that the statement is true for $n = k$, where k is a positive integer.
So assume

$$\sum_{r=1}^{k} r(r+1)(r+2) = \frac{1}{4}k(k+1)(k+2)(k+3)$$

Inductive Step: prove the statement is true for $n = k+1$
Starting with the left-hand side for $n = k+1$,

$$\sum_{r=1}^{k+1} r(r+1)(r+2)$$

Write this as

$$\sum_{r=1}^{k} r(r+1)(r+2) + (k+1)(k+2)(k+3)$$

Now use the inductive hypothesis:

$$\sum_{r=1}^{k+1} r(r+1)(r+2) = \frac{1}{4}k(k+1)(k+2)(k+3) + (k+1)(k+2)(k+3)$$

Factor out $(k+1)(k+2)(k+3)$:

$$= (k+1)(k+2)(k+3)\left(\frac{k}{4} + 1\right)$$

Simplify the bracket:

$$\frac{k}{4} + 1 = \frac{k+4}{4}$$

So

$$\sum_{r=1}^{k+1} r(r+1)(r+2) = (k+1)(k+2)(k+3)\left(\frac{k+4}{4}\right)$$

Hence

$$\sum_{r=1}^{k+1} r(r+1)(r+2) = \frac{1}{4}(k+1)(k+2)(k+3)(k+4)$$

This is exactly the required formula with $n = k+1$.

Conclusion:
The statement is true for $n = 1$, and true for $n = k + 1$ whenever it is true for $n = k$.
Therefore, by mathematical induction,

$$\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

for all positive integers n.